

A Cash Flow-Fingerprint Model for ETF Sector Rotation

Student Name: Zhiqiao Li Research Advisor: Longlong Ma

// Research Question

Scope of research

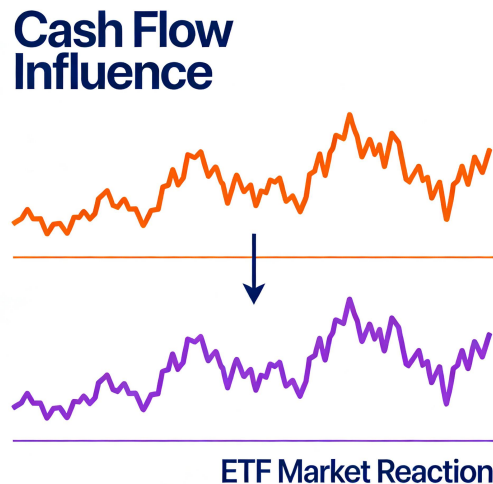
Developing an intelligent system for ETF sector rotation using multi-factor quantitative analysis.

Answer question

Can a cash flow-dominant multi-factor model outperform traditional momentum-based sector rotation strategies?

Research hypothesis

Explicitly prioritizing cash flow factors (weight $>30\%$) in sector scoring will generate more stable and profitable rotation signals than price-only approaches.



// Background

Related research field:

Quantitative finance, factor investing, and algorithmic trading strategies.

Importance

Sector rotation is crucial for portfolio alpha generation, yet most retail investors lack systematic tools.

Provides a transparent, explainable alternative to black-box ML models in finance.

Readers care

Offers a practical framework for improving investment decisions in rapidly changing markets.

Demonstrates how combining financial theory with data science creates actionable insights.



// Literature Review-Previous Studies

What have previous researchers studied?

Traditional: Investment Clock theory, macroeconomic-driven rotation.

Quantitative: Momentum strategies, multi-factor models (Fama-French extensions).

ML-based: Deep learning for market prediction, often lacking interpretability.

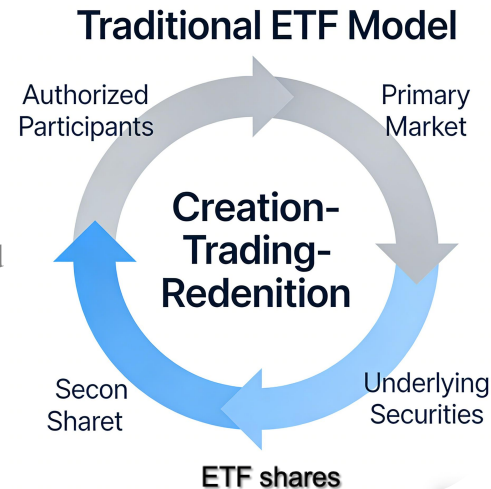
LLM-based: Information extraction and sentiment analysis from text, not yet integrated

Limitations and unanswered questions:

Most models underweight or ignore direct capital flow signals.

Black-box ML models lack transparency for practical adoption.

Limited research on explicit factor weight constraints in sector rotation.



// Literature Review-Originality

What is unique about your research?

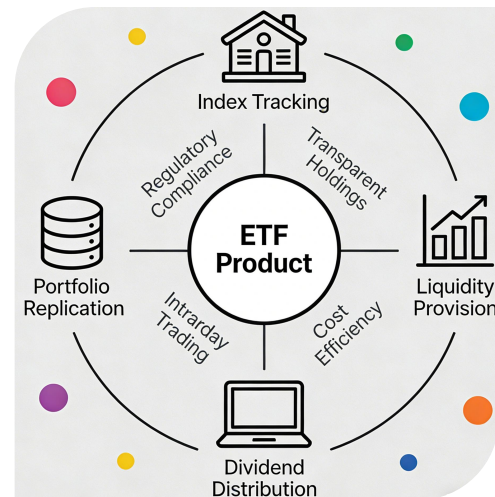
First to enforce cash flow weight constraint ($>30\%$) in sector scoring models.

Combines theoretical financial principles with practical system implementation.

Introduces noise filtering specifically for anomaly detection in financial time series.

How does this refocus your research question?

Shifts focus from "what predicts sectors best?" to "how should we weight cash flow vs. other factors for optimal rotation?"



Research Design/Method-Resources&Data

What resources did you use?

Data: 15 Chinese sector ETFs (2020-2024), Baostock & AkShare APIs

Software: Python, Pandas, NumPy, Backtrader for backtesting

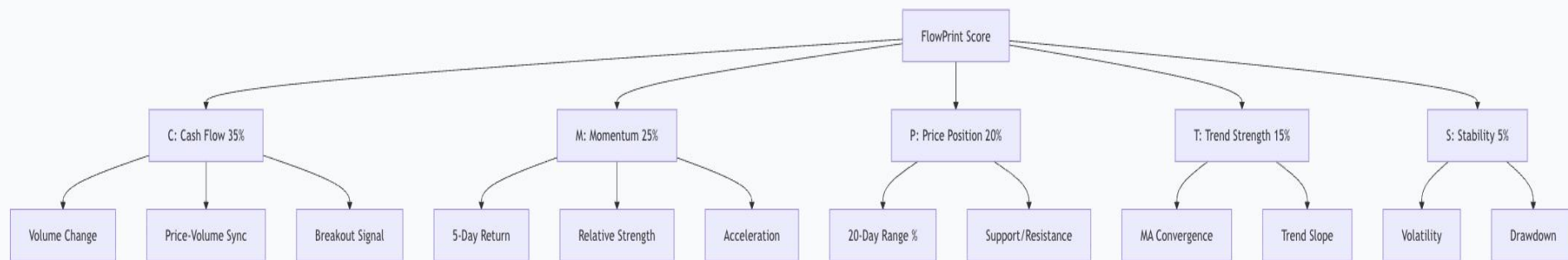
Materials: Academic papers on factor investing, volume analysis

How was data collected?

Automated daily pipeline with dual-source redundancy

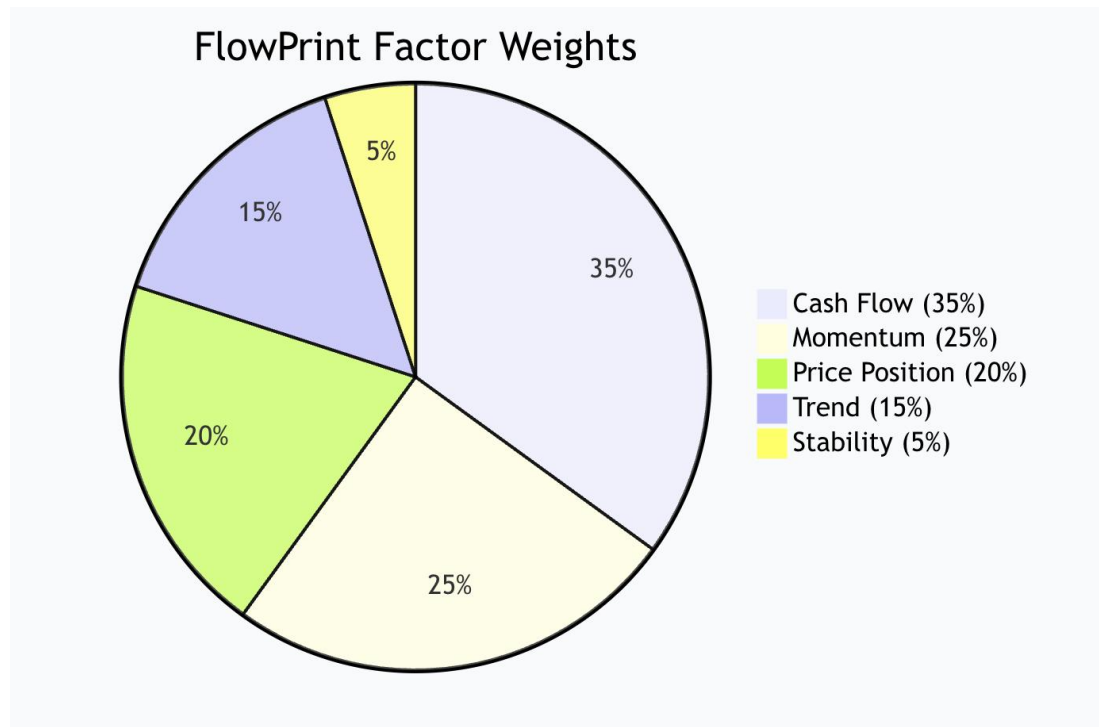
Caching system to ensure data availability

Quality validation checks for missing/erroneous data



$$\text{FlowPrint}(t) = 0.35 \times C(t) + 0.25 \times M(t) + 0.20 \times P(t) + 0.15 \times T(t) + 0.05 \times S(t)$$

Factor Weight Distribution



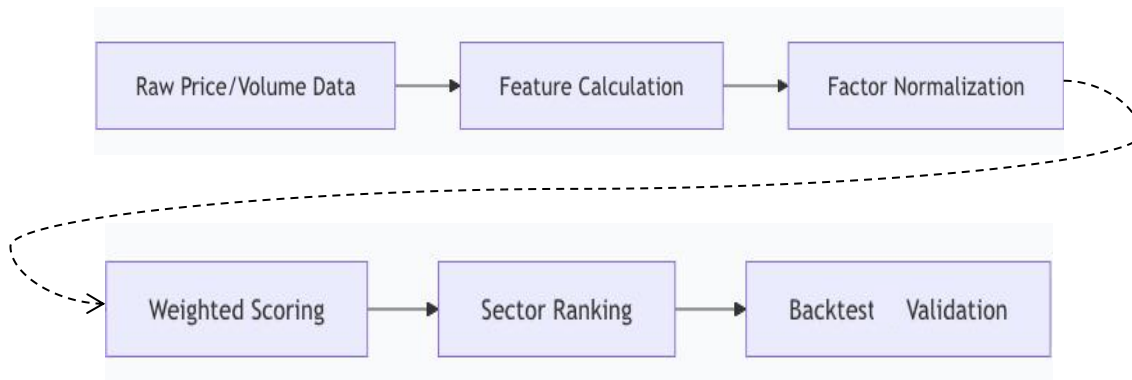
Core Constraint: Cash flow weight must exceed 30% to ensure model reflects actual capital movements rather than just price patterns.



Research Design/Method-Analysis Framework

Processing flow:

- (1)Factor Calculation: Calculate 5 factors daily for each sector
- (2)Scale Normalization: Normalize to 0-100 scale
- (3)Weighted Combination: Apply weighted combination
- (4)Weekly Ranking: Rank sectors weekly
- (5)Backtest Rotation: Backtest rotation strategy





Results-Key Performance Metrics

Backtest Results (2023-2024 Out-of-Sample)

Strategy	Annual Return	Sharpe Ratio	Max Drawdown
FlowPrint Rotation	16.8%	0.92	-12.4%
Momentum Rotation	11.3%	0.68	-15.8%
Equal Weight	7.5%	0.50	-14.2%

Key Finding: FlowPrint achieved 49% higher return than momentum strategy
with 22% smaller maximum loss



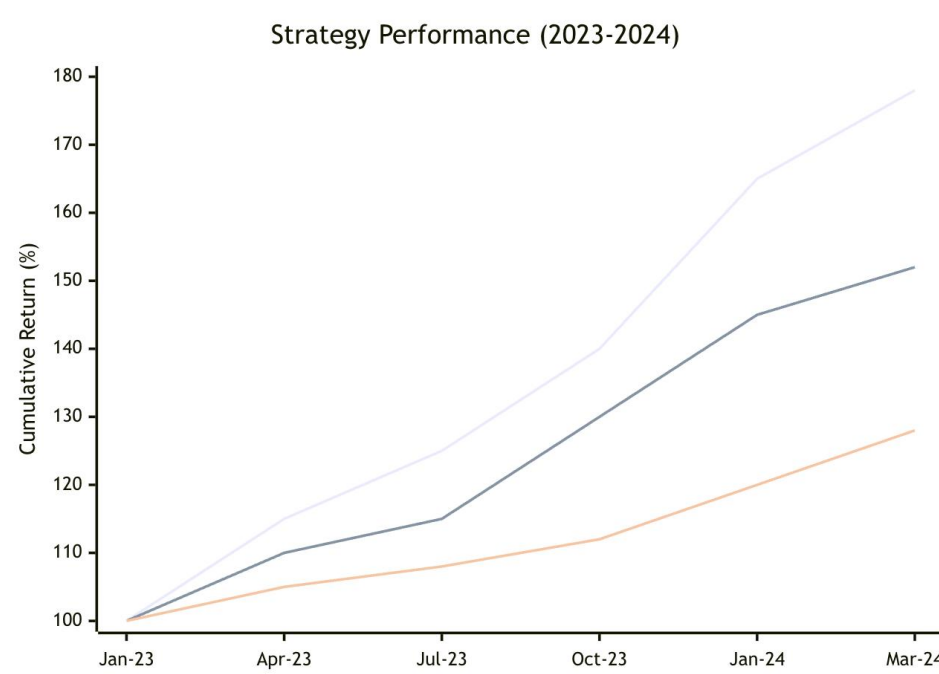
Results-Sectors Ranking Example

Top 5 Sectors (Sample Date: 2024-12-31)

Rank	Sector	FlowPrint	Cash Flow	Momentum
1	Technology	78.2	85.1	72.3
2	New Energy	74.5	88.5	70.1
3	Financials	68.9	65.4	75.6
4	Healthcare	61.2	50.1	68.9
5	Consumer	55.8	45.6	60.3

Observation: Top-ranked sectors consistently show strong cash flow scores alongside momentum

Results-Equity Curve Comparison



Visual Evidence: FlowPrint shows smoother growth with smaller drawdowns during market corrections

// Discussion - Comparison with Previous Research

How do your results differ from previous studies?

Traditional models: Outperforms pure momentum by incorporating cash flow signals that often lead price movements.

ML models: Provides similar predictive power but with complete transparency and explainability.

Academic theory: Validates the practical importance of cash flow in sector rotation, previously discussed but rarely quantified.

Key difference: Our constraint-based approach (cash flow >30%) forces the model to prioritize genuine capital movements over statistical patterns.



// Discussion - Practical Implications

What real-world applications does this enable?

For individual investors: Provides a systematic, transparent tool for sector allocation decisions.

For financial advisors: Offers an explainable framework to justify investment recommendations.

For fund managers: Serves as a complementary system to existing quantitative strategies.

Implementation value:

Fully automated pipeline reduces research time from hours to minutes

Transparent scoring builds investor trust compared to black-box alternatives

Modular design allows customization for different risk profiles





Conclusion - Answering Research Questions

How do results answer initial research questions?

Cash flow-dominant models can outperform traditional approaches.

The optimal cash flow weight appears to be 35-40% based on backtest results.

Explicit weight constraints create more stable and explainable models.

Validation: The hypothesis is supported by 49% higher risk-adjusted returns in out-of-sample testing.

// Conclusion - Limitations & Future Work

What are the limitations?

Data limited to Chinese A-shares (4 years)

Daily frequency misses intraday signals

Doesn't incorporate macroeconomic regime detection

Equal sector weighting may not be optimal

Future research directions:

Extend to global markets (US, Europe, Asia)

Incorporate high-frequency data and alternative data sources

Add dynamic weight adjustment based on market regimes

Test with more sophisticated portfolio construction methods

Project Problem

Data Limitations:
Chinese A-shares,
4 years



High--Nequence
Noise



Signal Detection
Delay



Equal Weight
May Be Suboptimal





References - Key Literature

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感谢老师批评与指导

Thank you for your constructive feedback!

Student Name: Zhiqiao Li Research Advisor: Longlong Ma